

Frontend Technical Task: Advanced Editable Data Table

Overview:

Build a data-intensive editable table component in React that supports:

- Editing numeric and text values
- Sorting, filtering, pagination
- Handling large datasets efficiently (10,000+ rows)

Requirements

1. Editable Table

- Users should be able to edit inline:
 - Text fields (e.g., name, email etc)
 - Numeric fields (e.g., salary, quantity)
- Provide save, cancel, and undo actions per row

2. Large Dataset Handling

- Handle a dataset of 10,000+ rows
- Implement performance optimizations:
 - Virtual scrolling (e.g., react-window, react-virtualized)
 - Pagination as fallback option

3. Sorting and Filtering(Optional)

- Multi-column sorting (asc/desc toggle)
- Text and numeric filtering
- Clear filters button

4. UI/UX Expectations

- Design should be based on you.
- Clean, accessible UI
- Use any UI library (e.g., Material UI, antd) or custom CSS

5. Bonus (Optional)

- Export table to CSV
- Track unsaved changes and prompt before leaving
- Implement a basic Redux or context-based state management for table data

Deliverables

- GitHub repo or zip with source code
- README with:
 - Setup instructions
 - Approach and decisions taken
 - Known limitations (if any)

Evaluation Criteria

- Code quality (modular, reusable, readable)
- Performance handling with large data
- State and form management
- UI/UX polish